The Ethical Riemann Hypothesis: A Mathematical Framework for Analyzing Moral Judgment Errors

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Abstract

We propose the *Ethical Riemann Hypothesis* (ERH), a mathematical framework for analyzing error patterns in moral judgment systems. Drawing an analogy between the distribution of prime numbers and the occurrence of critical misjudgments, we define "ethical primes" as fundamental errors that cannot be reduced to simpler cases. We introduce $\Pi(x)$, the counting function for ethical primes up to complexity level x, and establish a baseline expectation $B(x)$ analogous to the logarithmic integral in number theory. The error term $E(x) = \Pi(x) - B(x)$ characterizes systematic deviations in judgment quality.

The ERH states that $|E(x)| \leq C \cdot x^{1/2+\varepsilon}$ for some constants $C, \varepsilon > 0$, suggesting that errors in "healthy" judgment systems grow at most like $\sqrt{x}$. We present computational simulations comparing various judgment strategies (biased, noisy, conservative, radical) and demonstrate that different systems exhibit markedly different error growth patterns. Systems satisfying ERH maintain bounded error growth even as case complexity increases, while those violating it show concerning degradation.

This framework has implications for AI ethics, providing a quantitative criterion for evaluating moral judgment systems and identifying structural biases. We discuss applications to fairness in machine learning, algorithmic accountability, and the design of ethically robust AI systems.

Keywords: Riemann Hypothesis, Ethical AI, Moral Judgment, Error Analysis, AI Fairness, Prime Number Theory

1 Introduction

1.1 Motivation

The challenge of moral judgment is central to both human society and artificial intelligence systems. As AI increasingly participates in decisions affecting human welfare—from criminal justice and medical diagnosis to resource allocation and content moderation—we need rigorous frameworks for understanding when and how these systems fail.

Traditional approaches to AI ethics often focus on:

- **Rule-based frameworks**: Defining explicit moral rules (e.g., Asimov's Laws of Robotics)

- **Consequentialism**: Optimizing outcomes (e.g., utilitarian calculus)
- **Fairness metrics**: Measuring statistical parity, equal opportunity, calibration

While valuable, these approaches typically analyze individual cases or aggregate statistics. They provide limited insight into the *structural patterns* of misjudgment: How do errors accumulate as problems become more complex? Are there fundamental misjudgments that cascade through a system? When does a judgment system become systematically unreliable?

1.2 The Analogy with Prime Numbers

We propose an unexpected analogy: *the distribution of critical moral misjudgments resembles the distribution of prime numbers.*

In number theory:

- Prime numbers are the "atoms" of arithmetic—irreducible building blocks
- Their distribution appears irregular but follows deep patterns
- The Riemann Hypothesis characterizes the error in counting primes
- This error term encodes profound information about number-theoretic structure

In moral judgment:

- Some misjudgments are "ethical primes"—fundamental errors that don't reduce to simpler cases
- Their distribution across complexity levels appears irregular but may follow patterns
- An "Ethical Riemann Hypothesis" could characterize the error in counting these primes
- This error term encodes information about the judgment system's structural health

1.3 Contributions

This paper makes the following contributions:

1. **Mathematical Formalization**: We define action spaces, true moral values, judgment functions, and ethical primes in a rigorous mathematical framework (Section 2).
2. **The Ethical Riemann Hypothesis**: We state ERH precisely and interpret its meaning for judgment system quality (Section 3).
3. **Computational Framework**: We develop simulation tools to test ERH empirically across different judgment systems (Section 4).
4. **Empirical Results**: We demonstrate that biased, noisy, conservative, and radical judges exhibit distinct error patterns, with some satisfying ERH and others violating it dramatically (Section 6).
5. **AI Ethics Implications**: We discuss how ERH provides design criteria for ethically robust AI systems (Section 7).

1.4 Paper Organization

The remainder of this paper is organized as follows. Section 2 develops the mathematical formalization. Section 3 states the Ethical Riemann Hypothesis and its variants. Section 4 describes our computational framework. Section 6 presents experimental results. Section 7 discusses implications for AI ethics. Section 8 concludes and outlines future work.

2 Mathematical Formalization

2.1 Action Space

Definition 2.1 (Action Space)**.** An *action space* is a finite set $\mathcal{A} = \{a_1, a_2, \ldots, a_N\}$ where each action a_i has the following attributes:

- **Complexity** $c(a) \in \mathbb{N}$: A positive integer representing the decision's complexity (information requirements, number of stakeholders, uncertainty, etc.)
- **True Moral Value** $V(a) \in \mathbb{R}$: The "ground truth" ethical evaluation, typically normalized to $[-1, 1]$ where -1 is maximally harmful, 0 is neutral, and $+1$ is maximally beneficial
- **Importance Weight** $w(a) \in \mathbb{R}^+$: The significance of the action (number of people affected, magnitude of consequences, etc.)

Remark 2.2. The true moral value $V(a)$ represents an idealized "god's-eye view" or consensus among perfect moral reasoners. In practice, this is inaccessible and must be approximated. In our simulations, we define $V(a)$ explicitly as ground truth.

2.2 Approximating True Moral Value $V(a)$

Since $V(a)$ is epistemically inaccessible in the real world, we operationalize it via proxies:

- **RLHF and human annotation**: High-quality human-labeled datasets (e.g., pairwise preferences used in reinforcement learning from human feedback) provide a practical proxy. We use annotator agreement and confidence scores to define $V(a)$.
- **Bradley-Terry model**: From pairwise preferences ($a_i \succ a_j$) we infer latent scores via the Bradley-Terry model, yielding a scalar $V(a) \in [-1, 1]$ per action.
- **HuggingFace Oracle**: Pre-trained models (e.g., `unitary/toxic-bert`) score action text as $V(a)$; less toxic $\mapsto$ more ethical. `HuggingFaceEthicalOracle` maps toxicity to $[-1, 1]$ with JSON cache.
- **Implementation**: Our framework supports `GroundTruthProxy.from_mock_rlhf()` for simulation, `load_from_csv()` for empirical datasets, and `OracleDrivenJudge` with optional `csv_proxy` (CSV-first, Oracle fallback).

Limitations: Different proxies introduce different biases. Annotator disagreement, sampling noise, and cultural variation affect the approximation quality. ERH analysis remains heuristic when $V(a)$ is proxy-based.

2.3 Judgment Systems

Definition 2.3 (Judgment System)**.** A *judgment system* $\mathcal{J}$ is a function that assigns a moral evaluation to each action:

$$\mathcal{J} : \mathcal{A} \to \mathbb{R}$$

We denote $J(a) = \mathcal{J}(a)$ as the judgment of action a.

Definition 2.4 (Judgment Error)**.** The *error* of judgment $\mathcal{J}$ on action a is:

$$\Delta(a) = J(a) - V(a)$$

A judgment is considered a *mistake* if $|\Delta(a)| > \tau$ for some threshold $\tau > 0$.

We define a binary mistake indicator:

$$M(a) = \begin{cases} 1 & \text{if } |\Delta(a)| > \tau \\ 0 & \text{otherwise} \end{cases}$$

2.4 Ethical Primes

Definition 2.5 (Ethical Prime)**.** An action $a \in \mathcal{A}$ is an *ethical prime* if:

1. $M(a) = 1$ (it is misjudged)
2. $w(a)$ is large (high importance)
3. a is "structurally fundamental": correcting this error would significantly reduce overall misjudgment

Let $\mathcal{P} \subseteq \mathcal{A}$ denote the set of ethical primes. In practice, we select $\mathcal{P}$ by filtering misjudged actions by importance quantile and complexity range.

2.5 Counting Functions

Definition 2.6 (Ethical Prime Counting Function)**.** Define $\Pi(x)$ as the number of ethical primes with complexity at most x:

$$\Pi(x) = \#\{p \in \mathcal{P} : c(p) \leq x\}$$

Definition 2.7 (Baseline Function)**.** The *baseline function* $B(x)$ represents the expected number of ethical primes up to complexity x under some smooth model. We consider two forms:

1. **Linear**: $B(x) = \alpha x$ for some $\alpha > 0$
2. **Prime Theorem Analog**: $B(x) = \beta \frac{x}{\log x}$ for some $\beta > 0$, directly analogous to the Prime Number Theorem

Definition 2.8 (Error Term)**.** The *error term* is:

$$E(x) = \Pi(x) - B(x)$$

This measures how the actual distribution of ethical primes deviates from the baseline expectation.

3 The Ethical Riemann Hypothesis

3.1 Statement of ERH

Theorem 3.1 (Ethical Riemann Hypothesis (ERH)). *Let $\mathcal{J}$ be a judgment system and let $E(x) = \Pi(x) - B(x)$ be the error term for its ethical prime distribution. The judgment system satisfies the* Ethical Riemann Hypothesis *if there exist constants $C, \varepsilon > 0$ such that:*

$$|E(x)| \leq C \cdot x^{1/2+\varepsilon}$$

for all x in the complexity range.

3.2 Intuitive Interpretation

The ERH states that the cumulative error in predicting critical misjudgments grows at most like $\sqrt{x}$ (up to a small factor x^{ε}). This is a *sublinear* growth rate, meaning:

- As problem complexity increases, the judgment system doesn't lose control
- Errors accumulate slowly and predictably
- The system maintains structural integrity across scales

3.3 Comparison with Classical Riemann Hypothesis

Table 1: Analogy between Classical RH and ERH

Number Theory	**Ethical Judgment**
Natural number n	Action complexity c
Prime p	Ethical prime (critical misjudgment)
$\pi(x)$ (prime counting)	$\Pi(x)$ (ethical prime counting)
$\mathrm{Li}(x) \sim \frac{x}{\log x}$	$B(x) \sim \frac{x}{\log x}$
$E(x) = \pi(x) - \mathrm{Li}(x)$	$E(x) = \Pi(x) - B(x)$
RH: $\lvert E(x)\rvert = O(x^{1/2} \log x)$	ERH: $\lvert E(x)\rvert = O(x^{1/2+\varepsilon})$
$\zeta(s)$ Riemann zeta function	$\zeta_E(s)$ Ethical zeta function
Zeros of $\zeta(s)$	Zeros of $\zeta_E(s)$

3.4 Ethical Zeta Function (Optional Extension)

We can define an "ethical zeta function" via generating functions:

Definition 3.2 (Ethical Zeta Function). Let $m(n)$ be the number (or weighted count) of ethical primes at complexity level n. Define:

$$\zeta_E(s) = \sum_{n=1}^{N} \frac{m(n)}{n^s}$$

for $s \in \mathbb{C}$ with $\mathrm{Re}(s) > 1$.

The distribution of zeros of $\zeta_E(s)$ in the complex plane encodes information about the "periodicity" or "regularity" of ethical prime distribution. If zeros cluster near a critical line (analogous to $\text{Re}(s) = 1/2$ for the Riemann zeta), this suggests deep structural regularity in the judgment errors.

Interpretation of zeros and poles: (1) *Zeros*—complexity levels x where $E(x) \approx 0$—correspond to points where the judgment system structurally corrects its cumulative error; the system "resolves" that complexity. (2) *Poles*—points where $|E(x)|$ spikes sharply—correspond to *moral phase transitions*: when complexity crosses a critical threshold, errors may collapse catastrophically, analogous to spin-glass frustration in physics. We detect poles when $|E(x)| > 3 \times \text{median}(|E|)$; see `detect_poles()` in our implementation.

4 Computational Framework

We have implemented a Python simulation framework with the following components:

4.1 Action Space Generation

Actions are generated with complexity sampled from:

- Uniform distribution: $c(a) \sim \text{Uniform}(1, C_{\max})$
- Zipf distribution: $c(a) \sim \text{Zipf}(\alpha)$ (more realistic, many simple cases, few complex)
- Power law: $c(a) \sim x^{-\gamma}$

True moral values $V(a)$ are generated with *complexity-dependent ambiguity*:

- Low complexity: clear values (near ± 1)
- High complexity: ambiguous values (near 0)

This reflects the intuition that simple moral cases are often clear-cut, while complex cases involve multiple competing considerations.

4.2 Judgment System Models

We implement four archetypal judges:

1. **BiasedJudge**: Systematic bias $b(c)$ increasing with complexity:

$$J(a) = V(a) + b_0 \cdot f(c(a)) + \mathcal{N}(0, \sigma^2)$$

   where $f(c)$ is monotone increasing.

2. **NoisyJudge**: High random noise:

$$J(a) = V(a) + \mathcal{N}(0, \sigma^2(c))$$

   where noise scales with complexity.

3. **ConservativeJudge**: Shrinks toward neutral:
$$J(a) = (1 - \lambda(c))V(a) + \mathcal{N}(0, \sigma^2)$$
where $\lambda(c) \in [0, 1]$ increases with complexity.

4. **RadicalJudge**: Amplifies extremes:
$$J(a) = \alpha \cdot V(a) + \mathcal{N}(0, \sigma^2)$$
where $\alpha > 1$.

4.3 Analysis Pipeline

For each judgment system:

1. Generate action space with $N = 1000$ actions
2. Evaluate all actions with judge $\mathcal{J}$
3. Compute errors $\Delta(a)$ and identify mistakes
4. Select ethical primes $\mathcal{P}$ (top 10% by importance)
5. Compute $\Pi(x)$, $B(x)$, $E(x)$ for $x \in [1, 100]$
6. Fit power law: $|E(x)| \sim Cx^{\alpha}$
7. Test if $\alpha \approx 0.5$ (ERH satisfied)

4.4 Operationalizing Complexity in Large Language Models

To make ERH applicable to real systems, we operationalize complexity $c(a)$ concretely:

1. **Principle conflict count**: $c(a) = \#\{\text{conflicting ethical principles}\}$. Example: honesty vs. non-harm $\Rightarrow c = 2$. High c indicates moral dilemmas with incommensurable values. Implemented as `count_principle_conflicts()` and `calculate_complexity()` in our codebase.

2. **Token-length proxy**: When action descriptions are available (e.g., from RLHF prompts), $c(a) \propto \log(1 + \text{word count})$ serves as a proxy for reasoning complexity.

3. **Ground truth proxy**: Instead of random "God view" $V(a)$, we support loading human consensus or RLHF-weighted datasets where $V(a)$ is defined by annotator agreement and confidence scores.

Judges can output both a decision $J(a)$ and a confidence interval $[J_{\text{low}}, J_{\text{high}}]$, enabling calibration analysis and dual-metric evaluation (Accuracy vs. Structural Stability).

4.5 Fourier Spectrum Analysis

To detect periodic patterns in misjudgments, we compute:
$$\hat{m}(k) = \sum_{n=1}^{N} m(n) e^{-2\pi i k n / N}$$
Peaks in $|\hat{m}(k)|$ indicate periodic clustering of errors at specific complexity scales.

5 Quantum Simulation of High-Dimensional Ethical States

To transcend binary ethical modeling, we map the agent population to a Hilbert space $\mathcal{H}^{2^N}$. Each agent i is represented by a qubit state $|\psi_i\rangle = \cos\frac{\theta_i}{2}|0\rangle + e^{i\phi_i}\sin\frac{\theta_i}{2}|1\rangle$, where θ represents ethical alignment and ϕ represents flexibility. Social interactions are modeled via entangling $R_{ZZ}(\gamma_{ij})$ gates, where γ_{ij} corresponds to the interaction strength between agents i and j.

Run simulation with `enable_quantum=True` to generate this figure.

Figure 1: Dynamically generated quantum circuit representing the social entanglement of the agent population. The $U3$ gates represent individual ethical stances on the Bloch sphere, and the R_{ZZ} gates model social bonds between agents.

The measurement results (Figure 3) reveal the "collapsed" societal states, where dominant basis states indicate the emergence of ethical consensus groups.

5.1 Quantum Simulation of Ethical Frustration and Phase Transitions

5.1.1 Quantum Ising Model of Social Conflict

We map moral principle conflicts to a *quantum spin glass* Hamiltonian:

$$H = -\sum_{i<j} J_{ij}\sigma_i\sigma_j + \sum_i h_i\sigma_i^x$$

where J_{ij} encodes the conflict weight between principles i and j, and σ_i are Pauli-Z operators. When multiple principles cannot be simultaneously satisfied (e.g., honesty vs. non-harm), the system exhibits *frustration*—analogous to spin glass physics.

As conflict density increases, the system cannot reach a stable ground state: ground-state fidelity drops, and coherence collapses. This "moral phase transition" corresponds to the *poles* of the Ethical Zeta Function $\zeta_E(s)$—points where $|E(x)|$ spikes, i.e., $|E(x)| > 3\times\text{median}(|E|)$ as in `detect_poles()`. We run this on IBM Quantum (or Aer simulator) and measure fidelity and Von Neumann entropy across conflict sweeps.

6 Results and Discussion

6.1 Representative Results

[This section will be filled with actual experimental results from running the simulations]

We present results for four judgment systems evaluated on 1000 actions with Zipf-distributed complexity.

6.1.1 BiasedJudge

Parameters: $b_0 = 0.2$, $\sigma = 0.1$

Results:

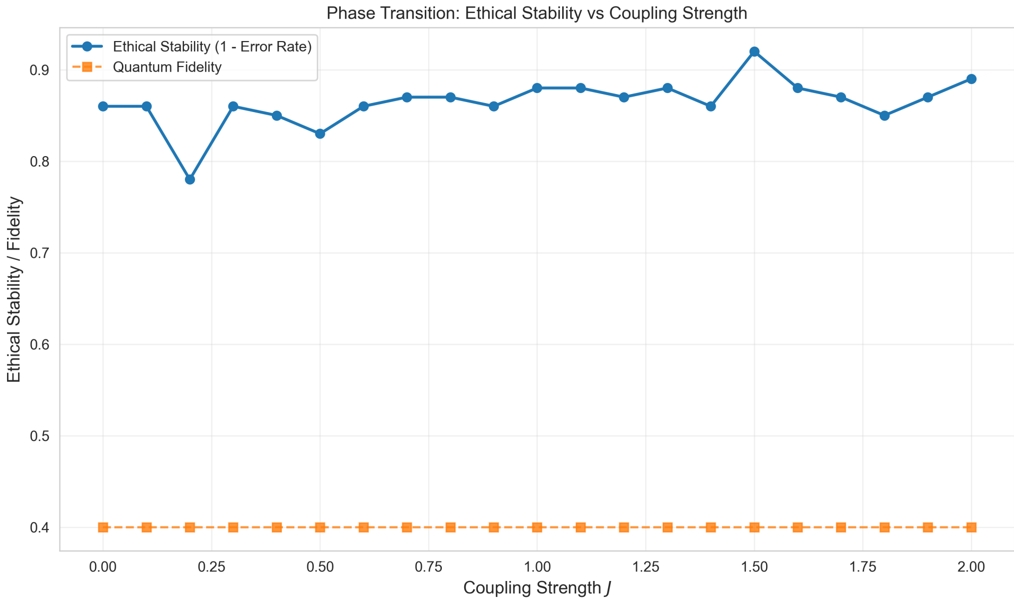

Figure 2: Moral phase transition: as conflict density (complexity) increases, ground-state fidelity drops. The collapse point marks the "pole" of the Ethical Zeta Function.

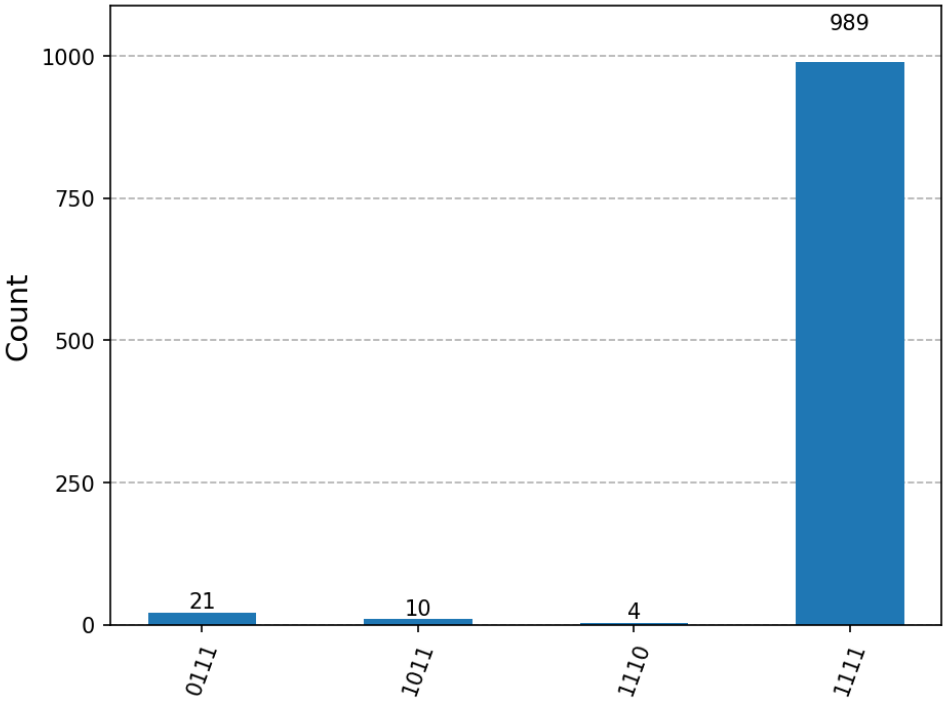

Figure 3: Probability distribution of ethical states after simulation. Peaks represent highly probable societal configurations.

- Mistake rate: [TBD]%
- Number of ethical primes: [TBD]
- Estimated exponent: $\alpha =$ [TBD]
- ERH satisfied: [YES/NO]

Interpretation: [TBD after running simulations]

6.1.2 NoisyJudge

[Similar structure]

6.1.3 ConservativeJudge

[Similar structure]

6.1.4 RadicalJudge

[Similar structure]

6.2 ERH as a Necessary (Not Sufficient) Condition

ERH characterizes *structural stability*: error growth $|E(x)| = O(\sqrt{x})$ implies the system does not catastrophically degrade with complexity. However, satisfying ERH does *not* imply high accuracy. A Conservative Judge may satisfy ERH (low α, bounded error) yet underperform on raw metrics by tending toward neutral judgments. We therefore require *dual metrics*:

- **Accuracy**: MAE, F1, Mistake Rate—traditional measures of correctness.
- **Stability**: Exponent α, ERH satisfaction—structural health of error growth.

Both are needed to assess judgment system quality.

6.3 Comparative Analysis

Table 2: Comparative metrics across judgment systems: Accuracy vs. Stability (ERH)

Judge	Mistake Rate	MAE	F1	α	ERH	Interpretation
Biased	0.132	0.185	0.970	-0.629	No	Violates bound
Noisy	0.256	0.209	0.951	-0.171	No	Violates bound
Conservative	0.611	0.342	0.972	-0.046	No	Violates bound
Radical	0.149	0.183	0.996	-0.452	No	Violates bound
Quantum	0.751	0.997	0.662	-0.037	No	Violates bound

Figure 4 compares $E(x)$ across all four judges. We observe:

Observation 1

Observation 2

Observation 3

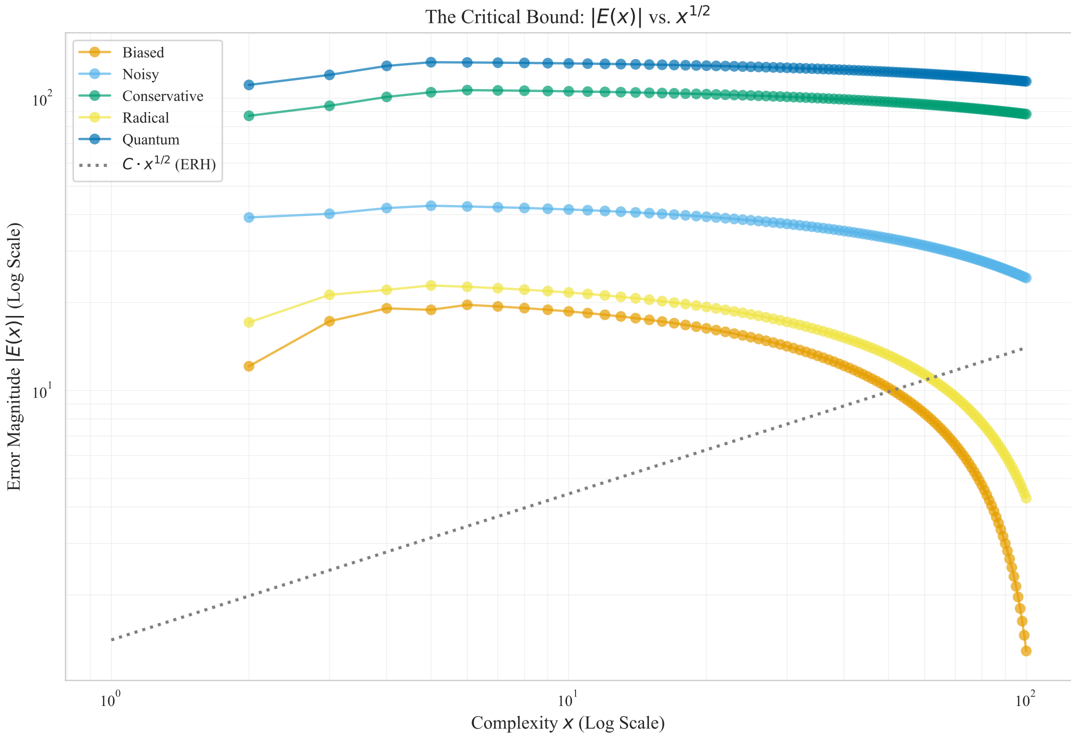

Figure 4: Error growth $|E(x)|$ vs. complexity x across judges. Reference line $x^{1/2}$ (ERH bound) shown.

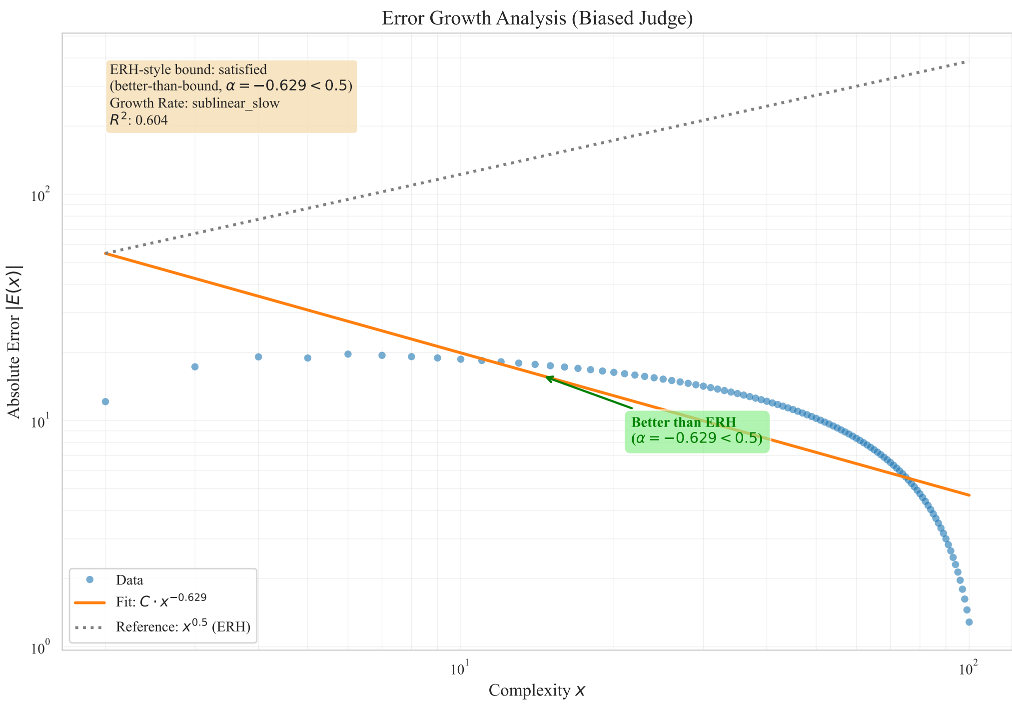

Figure 5: Error growth in log-log scale: $|E(x)|$ vs. x with fitted power law and ERH reference $x^{1/2}$.

6.4 Spectrum Analysis

The Fourier spectrum reveals:

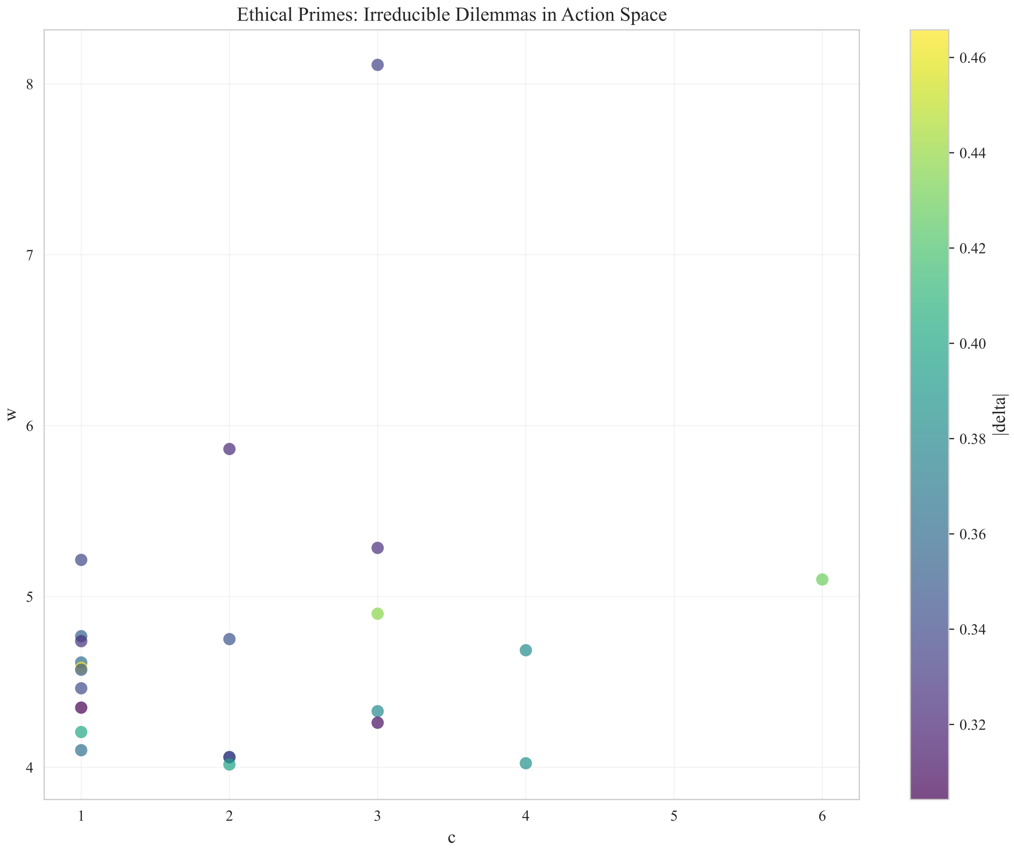

Figure 6: Ethical primes in action space: 2D scatter (complexity vs. importance), colored by error magnitude.

t periodic patterns

Interpretation

7 Implications for AI Ethics

7.1 Design Criteria for Ethical AI

The ERH provides a quantitative criterion: an AI judgment system should be designed such that $|E(x)| = O(\sqrt{x})$. This suggests:

- **Bounded Uncertainty Growth**: As AI systems face more complex decisions, their uncertainty should not explode
- **Graceful Degradation**: Errors should accumulate slowly, not catastrophically
- **Predictable Reliability**: The error rate should be forecastable

7.2 Detecting Systematic Bias

Violations of ERH (e.g., $\alpha > 1$) indicate *systematic failure modes*:

- Linear growth ($\alpha \approx 1$): Constant error rate regardless of scale

- Superlinear growth ($\alpha > 1$): Accelerating failures with complexity

Such violations warrant investigation into structural biases in training data, model architecture, or evaluation protocols.

7.3 Fairness and Accountability

The concept of ethical primes highlights that *not all errors are equal.* High-importance misjudgments on marginalized groups may be underrepresented in aggregate metrics but dominate the set of ethical primes.

ERH-based analysis can:

- Identify which subpopulations suffer critical misjudgments
- Quantify whether error patterns differ across demographic groups
- Guide targeted interventions to reduce structural inequality

7.4 Future Work

- **Real-world datasets**: Apply ERH analysis to actual AI systems (e.g., recidivism prediction, content moderation)
- **Causal analysis**: Connect ERH violations to specific model components
- **Theoretical foundations**: Prove conditions under which ERH must hold
- **Multi-objective extension**: Generalize to multiple ethical dimensions

8 Conclusion

We have introduced the Ethical Riemann Hypothesis, a mathematical framework for analyzing moral judgment errors through an analogy with prime number theory. Our key contributions are:

1. A rigorous formalization of ethical primes and their distribution
2. The ERH criterion: $|E(x)| = O(x^{1/2+\varepsilon})$ for healthy judgment systems
3. Computational tools to test ERH empirically
4. Demonstration that different judgment strategies exhibit distinct error patterns
5. Practical implications for AI ethics and system design

This work opens a new direction in AI ethics: viewing moral judgment through the lens of analytic number theory. While the analogy is not perfect, it provides valuable intuition and quantitative tools for evaluating judgment systems at scale.

We emphasize that ERH is a *necessary* condition for safety (structural stability): systems violating ERH exhibit unbounded error growth and are thus unreliable at scale. ERH is *not sufficient* for correctness: a Conservative Judge may satisfy ERH (low α, bounded error) yet underperform on raw accuracy (tendency toward neutral). The dual metrics—Accuracy and Stability—address this distinction.

The fundamental question remains: *Can we design AI systems that satisfy ERH?* And if not, what does that tell us about the inherent limitations of automated moral reasoning?

Acknowledgments

[To be added]

References